

Case 2: $y=1$. $n=10x+1$ $\therefore n^4=(10x+1)^4=(100x^2+20x+1)^2$
 $=10000x^4+4000x^3+4000x^2+40x+400x^2$
 $=10(1000x^4+400x^3+400x^2+4x+40x^2)+1$

$\hookrightarrow$ ends with 0, since it's multiplied by 10.

$\therefore n^4$ ends with 1 $\square$

Case 3: $y=2$. $n=10x+2$ $\therefore n^4=(10x+2)^4=(100x^2+40x+4)^2$
 $=10000x^4+1600x^3+16+8000x^2+320x+800x^2$
 $=10(1000x^4+160x^3+800x^2+32x+80x^2)+16$

$\hookrightarrow$ ends with 0, since it's a multiple of 10. $\therefore n^4$ ends with 6 $\square$

Case 4: $y=3$. $n=10x+3$
 $\therefore n^4=(10x+3)^4=(100x^2+60x+9)^2=10000x^4+3600x^3+81+12000x^2$
 $+1080x+1800x^2$
 $=10(1000x^4+360x^3+1200x^2+108x+180x^2)+81$

$\hookrightarrow$ ends with 0, since it's a multiple of 10. $\therefore n^4$ ends with 1 $\square$

Case 5: $y=4$. $n=10x+4$ $n^4=(10x+4)^4=(100x^2+80x+16)^2$
 $=10000x^4+6400x^3+256+16000x^2+2560x+3200x^2$
 $=10(1000x^4+640x^3+1600x^2+256x+320x^2)+256$

$\hookrightarrow$ ends with 0, since it's a multiple of 10.

$\therefore n^4$ ends with 6 $\square$

Case 6: $y=5$. $n=10x+5$ $n^4=(10x+5)^4=(100x^2+50x+25)^2$
 $=10000x^4+2500x^3+625+10000x^2+2500x+5000x^2$
 $=10(1000x^4+250x^3+100x^2+250x+500x^2)+625$

$\hookrightarrow$ ends with 0, since it's a multiple of 10. $\therefore n^4$ ends with 5 $\square$